PROBLEM STATEMENT #20 | Healthcare & Telemedicine

Hospital Bed & ICU Allocation Dashboard

1. Problem Context & Overview

An emergency room bed management dashboard displaying real-time ward occupancy, prioritizing ICU transfers based on triage score, and orchestrating inter-facility ambulance transfer workflows.

Target Stakeholders / Actors: *Triage Nurse, Hospital Administrator*

2. Sample Functional Requirement (FR) Guideline

Sample Requirement: FR-001 [Priority: High]
Description: The system shall compute triage severity scores from intake vitals and dynamically order the ER bed allocation waitlist.
Sample Acceptance Criteria: Pass: Higher acuity patient automatically moves ahead of lower acuity patient; Fail: FIFO queue ignores critical emergency.

3. Sample Non-Functional Requirement (NFR) Guideline

Sample Requirement: NFR-001 [Type: Performance & Security]
Description: Bed status transitions (Occupied, Cleaning, Available) must propagate to all connected ER dashboard clients in under 1 second via WebSockets.
Sample Acceptance Criteria: Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.

Deliverables to Submit: [All through a GitHub Repo in your account]

- **Complete Requirements Table:**
  - >> Exactly 5 Functional Requirements (FR-001 to FR-005)
  - >> 2 Non-Functional Requirements (NFR-001 & NFR-002) with
    ID, Type, Description, Priority, Acceptance Criteria, and Rationale.
- **UML Use-Case Diagram:**
  - >> Model all actors, primary use cases, and include at least one «include» and one «extend» relationship.
- **Use-Case Flow Specification:**
  - >> 1-page document for one core use case detailing Preconditions, Postconditions, Main Success Scenario
  - >> One Alternate Flow.